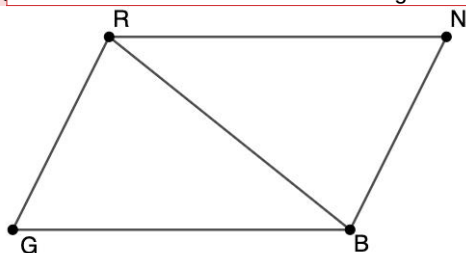


Geometry
Unit 8
Lesson 1 Homework

Name **Answer Key** _____
Date _____
Period _____

Quadrilateral $RNBG$ is shown with diagonal $\overline{RB}$.



Given: $\overline{RN} \cong \overline{GB}$, $\angle NRB \cong \angle GBR$

Prove: $RNBG$ is a parallelogram.

Commented [DB1]: <https://www.geogebra.org/calculator/myhffg3r>

Statement	Reason
$\overline{RN} \cong \overline{GB}$	Given
1. $\overline{RB} \cong \overline{RB}$	2. Reflexive Property of Congruence
$\angle NRB \cong \angle GBR$	Given
$\triangle RGB \cong \triangle RNB$	3. SAS $\cong$
4. $\overline{RN} \parallel \overline{GB}$	If two lines are intersected by a transversal and the alternate interior angles are congruent, then the lines are parallel.
5. $\angle BRG \cong \angle RBN$	6. Corresponding Parts of Congruent Triangles are Congruent
$\overline{RG} \parallel \overline{NB}$	If two lines are intersected by a transversal and the alternate interior angles are congruent, then the lines are parallel.
$ABCD$ is a parallelogram	7. A parallelogram has two sets of opposite parallel sides.

Use the statements and reasons provided to complete the two-column proof.

$\angle BRG \cong \angle RBN$	Corresponding Parts of Congruent Triangles are Congruent
$\angle RGB \cong \angle NBR$	Reflexive Property of Congruence
$\overline{RB} \cong \overline{RB}$	Transitive Property of Congruence
$\overline{RG} \cong \overline{NB}$	SSS $\cong$
$\overline{RN} \parallel \overline{GB}$	SAS $\cong$
$\overline{RG} \parallel \overline{NB}$	A parallelogram has two sets of opposite parallel sides.
	A parallelogram has two sets of opposite congruent sides.

Fill in the blanks with the correct terms to complete the statements.

8. A quadrilateral containing __two__ set(s) of ____parallel/congruent_ sides is a parallelogram.
9. __Opposite_____ interior angles of a parallelogram are __congruent_____.
10. The __diagonals_____ of a parallelogram bisect each other.